Unit ADC

SKU:U013










Description

Unit ADC is an A/D converter with a built-in 16-bit self-calibrating analog-to-digital converter **ADS1100**. Through the I2C communication protocol, the ADS1100 can perform conversions at 8, 16, 32, or 128 samples per second. The on-chip programmable gain amplifier (PGA) provides up to 8x gain. For applications requiring high-resolution A/D conversion acquisition, **Unit ADC** is the perfect solution, with an I2C address of 0x48.

Features

- Complete data acquisition system
- Package: TINY SOT23-6
- 16-bit no missing codes
- Continuous self-calibration
- Single-cycle conversion
- Internal system clock
- I2C interface
- 2 x LEGO compatible holes

Includes

- 1 x Unit ADC
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x HT3.96 Male Socket (2 pins)

Applications

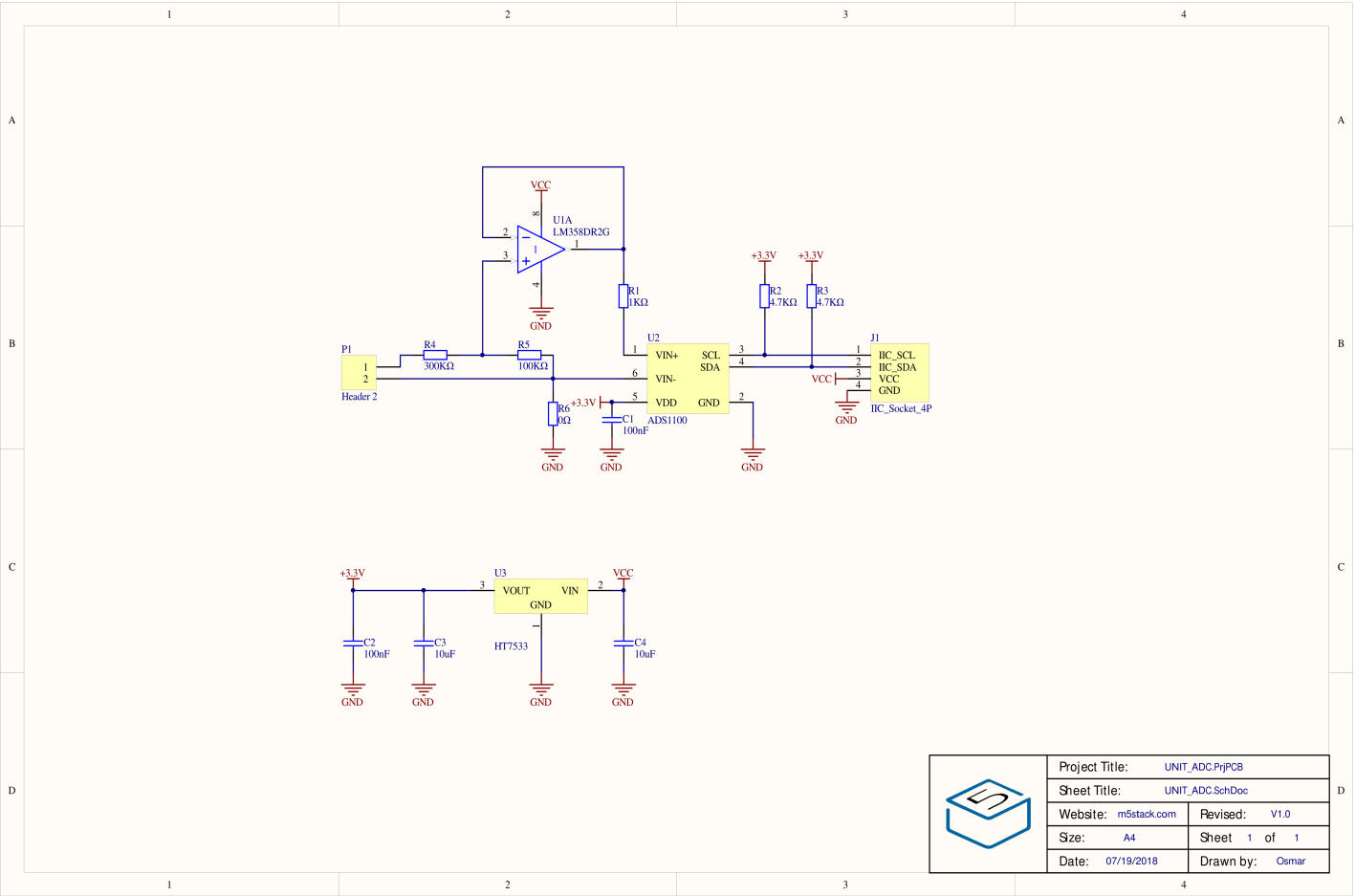
- ECG signal acquisition
- Blood pressure measurement
- Force gauge

Specifications

Specification	Parameter
INL	0.0125% of full-scale range (max)
Gain	1, 2, 4, 8
Programmable Rate	8SPS to 128SPS
Current	90μA
Noise	4μVp-p
Communication Interface	I2C @0x48
Product Size	32.0 x 24.0 x 10.2mm
Product Weight	4.2g
Package Size	138.0 x 93.0 x 11.3mm
Gross Weight	14.7g

Schematics

- Unit ADC Schematics PDF

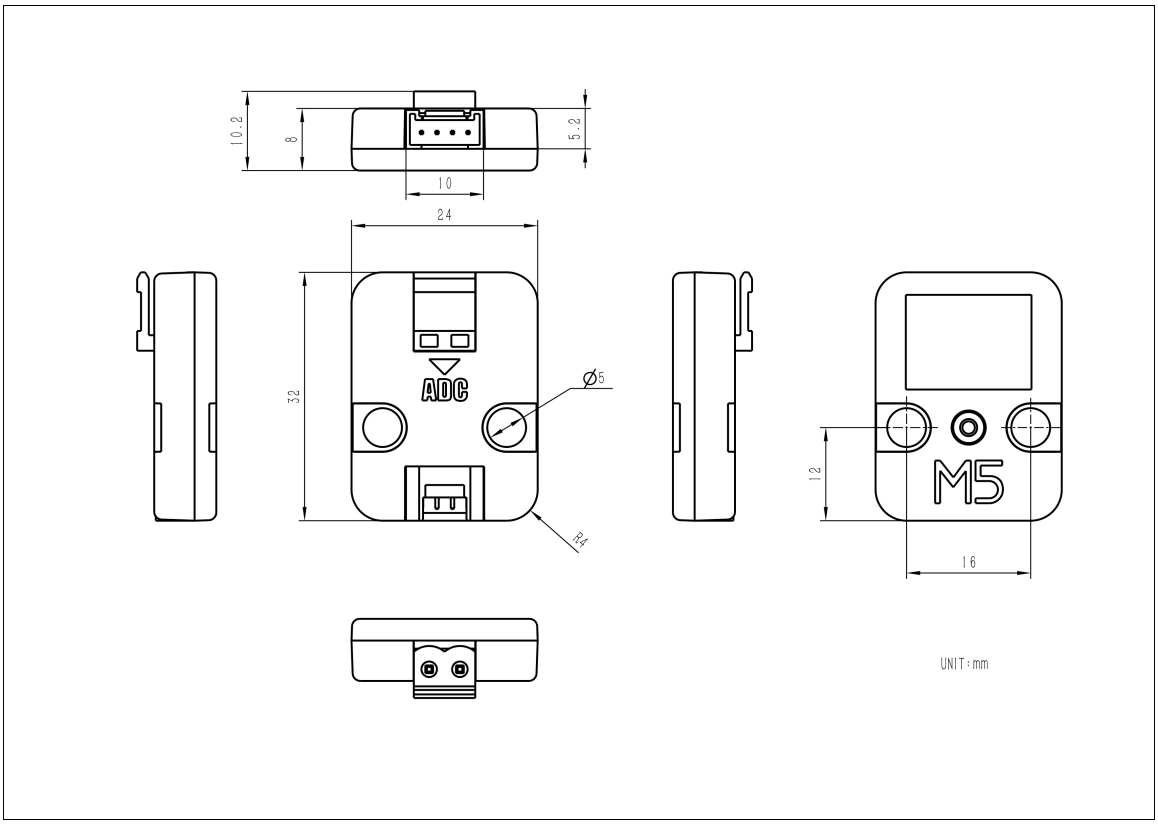


PinMap

Unit ADC

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Datasheets

- ADS1100

Softwares

Arduino

- Unit ADC with M5Core
- Unit ADC with M5Core2
- Unit ADC with M5Atom
- Unit ADC with M5StickC
- Unit ADC with M5StickCPlus

UiFlow1

- Unit ADC UiFlow1 Docs

UiFlow2

- Unit ADC UiFlow2 Docs

EasyLoader

Easyloader	Download Link	Note
Unit ADC example with M5Core	download	/

Video